

WASP-163b

R-2258

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-163_210513 · created 2026-08-16T00:53:20+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459347.8678 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.011
Transit depth (Rp/Rs)^2	1.89 +/- 0.31 [%]
Orbital Inclination (inc)	85.63 +/- 1.74
Airmass coefficient 1 (a1)	0.9044 +/- 0.0097
Airmass coefficient 2 (a2)	0.07 +/- 0.035
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	#1 - [555, 60]
Optimal Method	PSF photometry
Transit Duration (day)	0.0954 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.138 ± 0.011 vs 0.119 (deviation 1.73σ)
Duration fit vs published	0.0954 d vs 0.0942 d (deviation 0.18σ)
Mid-time O–C	2.5 min
Depth SNR	12.5
Residual scatter	1.07 %

Light curve

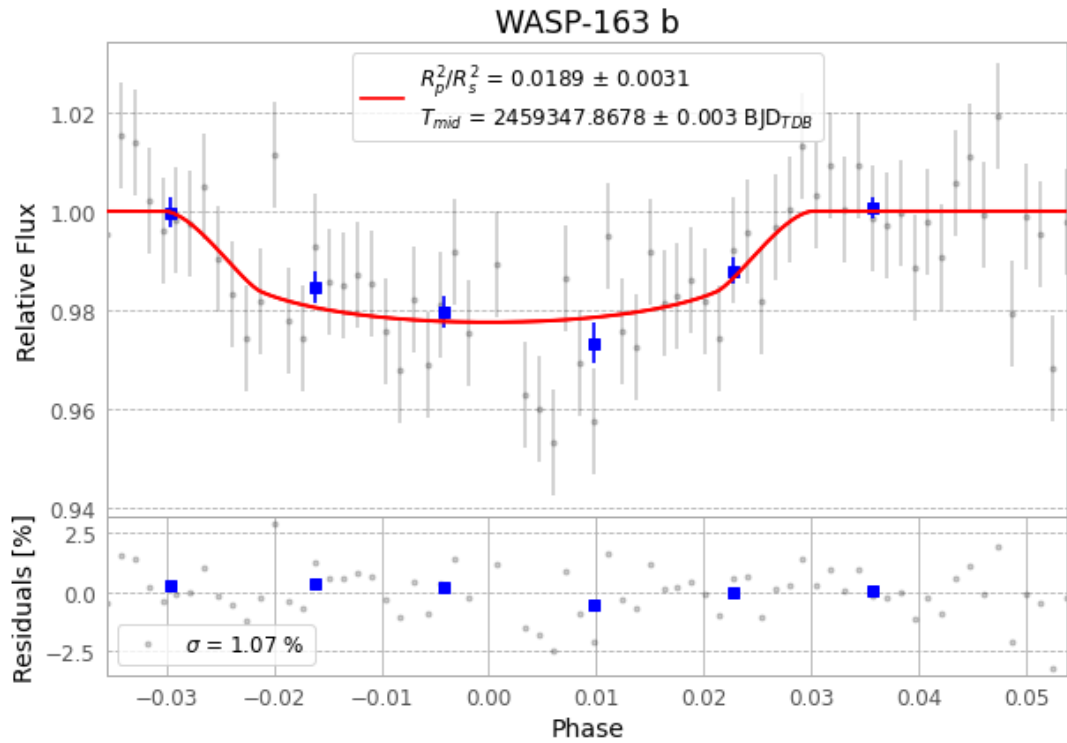


Figure 1 — FinalLightCurve_WASP-163 b_2021-05-13.png (SHA-256 cce27d15c5381663...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [399, 67], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0c65dadaa1f32a48...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

70 FITS frames (46 MB), WASP-163210513071812.FITS → WASP-163210513104512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-163-210513.log (11cace0e3253...), FinalLightCurve_WASP-163 b_2021-05-13.png (cce27d15c538...), FinalLightCurve_WASP-163 b_2021-05-13.csv (1049fb5e33fd...)

Compute resources

Wall time 97.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 00:53Z.